

Measure Theory and Probability: Worksheet 1

Chapter 1 — Lebesgue outer measure

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This worksheet contains the exercises corresponding to the associated lecture: Chapter 1 — Lebesgue outer measure. The exercises are deliberately ordered by difficulty: direct applications of the course, more demanding practice exercises, and then further theoretical material.

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1 Direct applications of the course

Exercise 1.1 – A part countable $\mathbb{R}$ is negligible

Show that if $D \subseteq \mathbb{R}$ is at most countable, then $\lambda^*(D) = 0$.

Exercise 1.2 – The Jordan content (comparison with the outer measure)

For $E \subseteq \mathbb{R}$, we define the *Jordan content* of E by

$$e(E) := \inf \left\{ \sum_{i=1}^n (b_i - a_i) \mid n \in \mathbb{N}, E \subseteq \bigcup_{i=1}^n [a_i, b_i] \right\},$$

the infimum being taken over all *finite* covers of E by closed bounded intervals (we compare it with $\lambda^*(E)$, which does not allow that of covers by intervals *open sets*, but in number *countable*).

1. Show that $e([a, b]) = b - a$ for all real $a \leq b$, and that e satisfies:

- (i) $e(\emptyset) = 0$;
- (ii) if $E \subseteq F \subseteq \mathbb{R}$, then $e(E) \leq e(F)$;
- (iii) for every family *finite* $(E_j)_{1 \leq j \leq m}$ of subsets of $\mathbb{R}$,

$$e\left(\bigcup_{j=1}^m E_j\right) \leq \sum_{j=1}^m e(E_j).$$

- 2. Show that $e(\overline{E}) = e(E)$ for all parts $E \subseteq \mathbb{R}$, where $\overline{E}$ denotes the closure (topological) of E .
- 3. The set function e can be identified to the outer measure of Lebesgue λ^* ?
- 4. The property (iii) above extends-it to the families *at most countable* of subsets of $\mathbb{R}$?

Exercise 1.3 – Approximation outer by a measurable set similarly measure

Show that for every $E \subseteq \mathbb{R}$, we have

$$\lambda^*(E) = \min\{\lambda^*(B) : B \in \mathcal{B}(\mathbb{R}), B \supseteq E\} = \min\{\lambda^*(A) : A \in \mathcal{M}(\lambda^*), A \supseteq E\}.$$

Exercise 1.4 – The square of a null set is negligible

Let $N \subseteq \mathbb{R}$ such that $\lambda^*(N) = 0$. Show that the set

$$N^2 := \{x^2 : x \in N\} \subseteq \mathbb{R}$$

satisfies $\lambda^*(N^2) = 0$.

Hint. Assume, for fix ideas, that we have $N \subseteq \mathbb{R}_+$. Show whereas, for each integer $n \in \mathbb{N}$, the set $(N \cap [0, n])^2$ satisfies $\lambda^*((N \cap [0, n])^2) = 0$.

Exercise 1.5 – Image of a null set by a function lipschitzienne

Recall that $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *lipschitzienne of constant* $L > 0$ if, for all $x, y \in \mathbb{R}$, we have:

$$|f(x) - f(y)| \leq L|x - y|.$$

1. Show that we have, for all parts $E \subseteq \mathbb{R}$:

$$\lambda^*(f(E)) \leq L \lambda^*(E).$$

2. Deduce that if $\lambda^*(N) = 0$, then $\lambda^*(f(N)) = 0$.

Exercise 1.6 – Characterization of the measurability by sandwichment open/closed

Show that $A \subseteq \mathbb{R}$ is Lebesgue-measurable if and only if, for all $\varepsilon > 0$, there exists an open U and a closed set F for which we have

$$F \subseteq A \subseteq U \quad \text{and} \quad \lambda^*(U \setminus F) \leq \varepsilon.$$

Exercise 1.7 – Continuity increasing of the outer measure

Let $(E_j)_{j \in \mathbb{N}}$ be an increasing sequence of subsets of $\mathbb{R}$. Show that

$$\lambda^*\left(\bigcup_{j \in \mathbb{N}} E_j\right) = \lim_{j \rightarrow \infty} \lambda^*(E_j).$$

Hint. Use Exercise 1.3 and the continuity increasing of a measure (course, Chapter 3), applied to the measure λ^* restricted to the sigma-algebra $\mathcal{M}(\lambda^*)$.

Exercise 1.8 – Preimage of a Borel sigma-algebra by a continuous function

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that if $B \in \mathcal{B}(\mathbb{R})$, then $f^{-1}(B) \in \mathcal{B}(\mathbb{R})$.

2 Practice exercises

Exercise 2.1 – Criterion of measurability via a test set of finite measure

We work with Lebesgue outer measure on $\mathbb{R}$. We are given $A \subseteq \mathbb{R}$ satisfying $\lambda^*(A) < +\infty$ and a set Lebesgue-measurable $X \subseteq \mathbb{R}$, containing A and satisfying $\lambda^*(X) < +\infty$. Show that then A is Lebesgue-measurable if and only if we have

$$\lambda^*(X) \geq \lambda^*(A) + \lambda^*(X \setminus A). \quad (1.19)$$

To show that the condition is sufficient, assume that the inequality (1.19) is satisfied, fix an arbitrary set $E \subseteq \mathbb{R}$ and set $F := E \cap X$.

1. Show that there exists a Lebesgue-measurable set B satisfying $F \subseteq B \subseteq X$ and $\lambda^*(B) = \lambda^*(F)$ (use the previous exercise).
2. Deduce from the Lebesgue-measurability of B that we have $\lambda^*(B \cap A) + \lambda^*(B \setminus A) \leq \lambda^*(B)$.
Hint. Observe first that

$$\lambda^*(B) + \lambda^*(X \setminus B) = \lambda^*(B \cap A) + \lambda^*(B \setminus A) + \lambda^*(A \setminus B) + \lambda^*(X \setminus (A \cup B)).$$

Deduce the undesired equality.

3. Deduce from which precedes that $\lambda^*(F \cap A) + \lambda^*(F \setminus A) \leq \lambda^*(F)$.
4. Conclude that $\lambda^*(E \cap A) + \lambda^*(E \setminus A) \leq \lambda^*(E)$.

Exercise 2.2 – The Vitali set is not measurable

Deduce from the previous exercise that the set $F \subseteq [0, 1]$ of Vitali (built in Chapter 1 of the course) satisfies $\lambda^*(F) + \lambda^*([0, 1] \setminus F) > 1$.

Exercise 2.3 – The Cantor devil staircase

We systematically use the notations in the course for the ternary Cantor set $C = \bigcap_{n \in \mathbb{N}} C_n$ (where $C_0 = [0, 1]$, and C_{n+1} is obtained by removing, from the middle of each interval making up C_n , its third central open). For each $n \in \mathbb{N}$, we define a function $\varphi_n : [0, 1] \rightarrow \mathbb{R}$ by the formula

$$\varphi_n(x) := \left(\frac{3}{2}\right)^n \lambda^*([0, x] \cap C_n),$$

for $x \in [0, 1]$.

1. Show that, for each $n \in \mathbb{N}$, φ_n is uniformly continuous on $[0, 1]$.
2. Show that the sequence $(\varphi_n)_{n \in \mathbb{N}}$ converge uniformly to a continuous function φ on $[0, 1]$.
3. Show that φ is increasing and satisfies $\varphi(0) = 0$, $\varphi(1) = 1$.
4. Show that $\varphi([0, 1] \setminus C)$ is at most countable; in deduce that $\lambda^*(\varphi(C)) = 1$.

We have therefore built a continuous function $\varphi : [0, 1] \rightarrow \mathbb{R}$ which maps a Lebesgue-null set (C) to a set of outer Lebesgue measure strictly positive; this function is often called the *staircase of the devil* of Cantor.

Exercise 2.4 – The complement of a null set is dense

Show that if $\lambda^*(A) = 0$, then A^c is dense in $\mathbb{R}$.

Exercise 2.5 – A part of measure half

Being given $E \subseteq \mathbb{R}$, show that there exists $F \subseteq E$ satisfying

$$\lambda^*(F) = \frac{1}{2} \lambda^*(E).$$

Show that if E is Lebesgue-measurable (respectively Borel sigma-algebra), then we can in outre choose F measurable (respectively Borel sigma-algebra).

Exercise 2.6 – Additivity of the outer measure for two sets at positive distance

Let $E, F \subseteq \mathbb{R}$ be arbitrary subsets satisfying

$$\text{dist}(E, F) := \inf\{|x - y| : x \in E, y \in F\} > 0.$$

Show that we have $\lambda^*(E \sqcup F) = \lambda^*(E) + \lambda^*(F)$.

Exercise 2.7 – The difference of the Cantor set with itself

We systematically use the notations in the course for the ternary Cantor set $C = \bigcap_{n \in \mathbb{N}} C_n$: C_n is a union of 2^n closed intervals pairwise disjoint, each of length 3^{-n} (we call them the *intervals of rank n*), and C_{n+1} is obtained by removing of each interval of rank n its third central open. We offer show the result following, to first seen surprenant since $\lambda(C) = 0$:

$$C - C := \{x - y : x, y \in C\} = [-1, 1].$$

1. Show that $C - C \subseteq [-1, 1]$.
2. (The not geometric key) Let $I = [a, a + \ell]$ and $J = [b, b + \ell]$ two closed intervals similarly length $\ell > 0$. We denote by $I^- := [a, a + \ell/3]$ and $I^+ := [a + 2\ell/3, a + \ell]$ the two under-intervals extreme obtained in reshoso many I its third central open (and similarly for J^-, J^+ to subset of J). Show that

$$\{y - x : x \in I, y \in J\} = \bigcup_{\epsilon, \eta \in \{-, +\}} \{y - x : x \in I^\epsilon, y \in J^\eta\}.$$

3. Deduce, by induction on $n \in \mathbb{N}$, that for every $\alpha \in [-1, 1]$, there exists a sequence $(I_n, J_n)_{n \in \mathbb{N}}$, where I_n and J_n are intervals of rank n (not necessarily distinct) making up C_n , satisfying $I_{n+1} \subseteq I_n, J_{n+1} \subseteq J_n$ for every n , as well as

$$\alpha \in \{y - x : x \in I_n, y \in J_n\} \quad \text{for every } n \in \mathbb{N}.$$

4. For each $n \in \mathbb{N}$, choose $(x_n, y_n) \in I_n \times J_n$ satisfying $y_n - x_n = \alpha$. Show that $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are of sequences of Cauchy in $[0, 1]$.
5. In notant $x := \lim_n x_n$ and $y := \lim_n y_n$, show that $x, y \in C$ and $y - x = \alpha$.
6. Conclude that $C - C = [-1, 1]$.

This result is frappant: although $\lambda(C) = 0$ (the set of Cantor is negligible), its difference with $C - C$ remplit *all* the segment $[-1, 1]$ — in particular, $D(C) := C - C$ contains all the rationals of $[-1, 1]$, and same all the irrationnels. Comparez with Exercise 1.18: the Sierpiński theorem showed that $D(A)$ contains a rational nonzero as soon as $\lambda^*(A) > 0$; the example of C shows that the converse is false — a set of outer measure zero can despite all have a set of differences also gros that possible.

Exercise 2.8 – Deviation of measure between a set not measurable and its parts measurable

Let $E \subseteq \mathbb{R}$ be a part not Lebesgue-measurable satisfying $\lambda^*(E) < +\infty$. Show that there exists $\eta > 0$ such that, for every subLebesgue-measurable set $A \subseteq E$, on ait $\lambda^*(A) \leq \lambda^*(E) - \eta$.

3 Further work

Exercise 3.1 – A set uncountable and negligible (via the Baire theorem)

Let us enumerate $\mathbb{Q}$ by the sequence $(q_k)_{k \in \mathbb{N}}$. For $\varepsilon > 0$, we define a set

$$U_\varepsilon := \bigcup_{k \in \mathbb{N}} \left[q_k - \frac{\varepsilon}{2^{k+2}}, q_k + \frac{\varepsilon}{2^{k+2}} \right]$$

we set then $G := \bigcap_{j \in \mathbb{N}} U_{2^{-j}}$.

1. Show that $\lambda^*(G) = 0$ and that $G \supseteq \mathbb{Q}$.
2. Show that, for every $\varepsilon > 0$, U_ε^c is of empty interior.
3. Deduce that G^c can be written as a countable union of closed sets of empty interior.
4. Conclude that G cannot be countable.

Hint. Proceed by contradiction and use the Baire theorem, which we recall here (classical result of topology, admitted in this course, in the same way that the Heine–Borel theorem):

Reminder (Baire theorem). In a space metric complete—in particular in $\mathbb{R}$ — every countable intersection of open sets dense is still dense. In a way equivalent, by passing to the complement: $\mathbb{R}$ cannot be written as a countable union of closed sets with empty interior.

We have therefore shown that there exists sets Lebesgue-negligible uncountable (different from $\mathbb{Q}$): there is strictly more sets Lebesgue-negligible than of countable parts $\mathbb{R}$. The set ternary Cantor set, seen in Chapter 1 of your course, provides another example.

Exercise 3.2 – A Lebesgue-measurable set non-Borel sigma-algebra

We keep the notation of the two previous exercises. We define a function continuous and strictly increasing $G : [0, 1] \rightarrow [0, 2]$ by setting, for $x \in [0, 1]$, $G(x) := \varphi(x) + x$.

1. Show that G is invertible and that its inverse $G^{-1} : [0, 2] \rightarrow [0, 1]$ is strictly increasing and continuous.
2. Show that if I is an interval content in $[0, 1] \setminus C$, then the interval $G(I) \subseteq [0, 2]$ satisfies $\lambda^*(G(I)) = \lambda^*(I)$.
3. Deduce that $\lambda^*(G([0, 1] \setminus C)) = 1$ and $\lambda^*(G(C)) = 1$.
4. Show that there exists a part E not Lebesgue-measurable of $\mathbb{R}$ contained in $G(C)$.
5. Observe that $N := G^{-1}(E)$ is of outer measure zero, therefore Lebesgue-measurable.
6. Show that on the other hand, N cannot belong to $\mathcal{B}(\mathbb{R})$, without what $E = (G^{-1})^{-1}(N)$ would be also Borel sigma-algebra (see Exercise 1.6).

Exercise 3.3 – A set not measurable is never approximated in measure by a measurable

Let $A \subseteq \mathbb{R}$ be a Lebesgue-measurable set satisfying $\lambda^*(A) > 0$. We propose to show that the set

$$D(A) := \{x - y : x, y \in A\} \subseteq \mathbb{R}$$

contains a rational nonzero. This is a result due to W. Sierpiński.

1. Show that it is enough to prove the result in the case where A is bounded.

Assume from now on A bounded, proceed by contradiction and assume that $D(A)$ does not contain no rational nonzero. Set still, for each $k \in \mathbb{N}$:

$$A_k := \frac{1}{2^k} + A := \left\{ \frac{1}{2^k} + x : x \in A \right\}.$$

2. Show that the sets A_k ($k \in \mathbb{N}$) are pairwise disjoint.
3. Deduce that $\lambda^*\left(\bigsqcup_{k \in \mathbb{N}} A_k\right) = +\infty$.
4. Conclude that $D(A)$ contains a rational nonzero.
5. Show that the Vitali set $F \subseteq [0, 1]$ in the course satisfies: $D(F)$ contains no rational nonzero. Comment this can-il?

Exercise 3.4 — Atteindre the outer measure by an under-measurable set implies the measurability

Let $E \subseteq \mathbb{R}$ satisfying $\lambda^*(E) < +\infty$. Show that if exists $A \subseteq E$ Lebesgue-measurable satisfying $\lambda^*(A) = \lambda^*(E)$, then E is Lebesgue-measurable.